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REVIEW



Free throw shooting performance under pressure: a social psychology critical review of research

Nadav Goldschmied^a, Mike Raphaeli^b, Samantha Moothart^a and Philip Furley^c

^aDepartment of Psychological Sciences, University of San Diego, San Diego, CA, USA; ^bNanning College for Vocational Technology, Nanning, China; ^cGerman Sport University Cologne, Cologne, Germany

ABSTRACT

The Free-Throw Shot (FTS) in basketball is a unique uncontested closed skill undertaken during a stoppage in play at a uniform distance. In recent years, FTS has emerged as a prime target for stress research due to its highly controlled nature and its significant impact on the game outcome. In this narrative review, we appraise empirical research with a social-psychological focus on studying FTS success under pressure. This critical review comprises 44 empirical studies published up to June 2020. We first assess the importance of FTS within the game of basketball and study historical trends. We then elaborate on the unique psychological features of FTS. Subsequently, research on the undermining effects of pressure from both archival game studies and experiments is critically evaluated. We conclude with a discussion of performance in relationship to expertise and sources of threat. The review highlights current research deficiencies, provides recommendations for theoretical research and applied practice.

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performance; pressure;
choking; expertise

On 3 April 1989 during the National Collegiate Athletic Association (NCAA) Championship game, a Michigan team trailed its opponent by 1 point with 3 s left in overtime. Michigan's point guard, Rumeal Robinson, was then fouled and stepped to the free throw shot (FTS) line. Robinson was averaging 57% in the tournament, below his already less than stellar 67% FTS career average. With more than 39,000 spectators in attendance, he made the two shots, giving Michigan its first and only National Championship. Nearly 30 years later, during Game 1 of the 2018 National Basketball Association (NBA) Finals, the visiting Cleveland Cavaliers trailed the Golden State Warriors by 1 point, with 4.7 s left in the game. George Hill, an 11-year league veteran and a career 80% FTS was fouled. He made his first attempt but missed the crucial second. His team went on to lose the game in overtime and subsequently the series. Anecdotal evidence, however memorable, naturally does not amount to scientific collection of observations. Henceforth we describe studies detailing the unique nature of the FTs, historical trends and its overall importance in basketball.

The free throw shot in basketball

In the game of basketball, the FTS stands unique. Generally, the game is characterised by its dynamic and interactive nature, which demands constant adjustment by players to their opponents' manoeuvring. However, the FTS differs fundamentally, as the player who is about to undertake it has neither significant time constraints (for example, the National Basketball Association [NBA] and Women's National Basketball Association [WNBA] allow an ample 10 s to execute), nor an opponent attempting to obstruct the attempt. It is performed from the same distance time and again (15-foot/4.60 m), in launching a 29.5" circumference ball weighing approximately 22 oz. or 28.5" circumference and 20 oz. in the women's game into an 18" diameter rim. This entirely closed type of athletic skill is thus controlled solely by the performer. NBA Hall-of-Famer, Rick Berry, a highly efficient FTS performer characterised it as: "the only part of the game that's is constant" (Fixler, 2012). Keetch et al. (2008) as well as Keetch et al. (2005) termed it an "especial skill" and conducted a series of studies corroborating the FTS uniqueness. They demonstrated that shooting accuracy in practice while being unguarded decreased linearly as shots were attempted from an increasing distance (9–21 feet in their study). However, when attempts were made from the foul line the success rate was higher than predicted from the regression line. Furthermore, FTS success was higher than other shoots made from the same distance but from different angles.

Branch (2009) in the popular press surveyed many years of American basketball and lamented the lack of improvement in FTS success which consistently hovered around 70% in the NCAA and 75% in the NBA [fluctuating quite narrowly between 72.8% in 1999 and 77.1% in 2010]. The statistic for the WNBA as well as the collegiate women have plateaued similarly. This trend is in remarkable contrast to other shooting attempts. For example, in men's college basketball, the overall field-goal percentage was below 40% until 1960, then rose gradually to 48.1 in 1984. The NBA long-range 3-point shot improved from 28% in the beginning of the 1980s to 37% in 2009. We extended this archival inquiry to the 2019 season (see Figure 1) and found still remarkable consistency (except for the WNBA FTS, which seems to improve) in contrast to field goal shooting (both 2 and 3 points attempts) in the NBA (see Figure 2).

The importance of the free throw shot on success in basketball

Before delving into the research exploring the psychological foundations of the shot, it is important to gauge the significance of the FTS in winning basketball games. How frequently is it attempted? What percentage of points made by a team stems from FTS? Does this unique shot become more important in a close game as the final outcome looms? Kozar et al. (1994) studied 490 National Collegiate Athletic Association (NCAA) division I games and found that about a fifth of all the points scored originated from FTS. This percentage more than doubled (45%~) in the last five minutes of the games for the winning team, while remaining about the same for the losers (21%~). This difference was even more pronounced in games decided by nine points or less. Similarly, Navarro et al. (2009) studied critical moments in the Spanish League in the last 5 min of close games and identified successful FTS as statistically discriminating between winning and

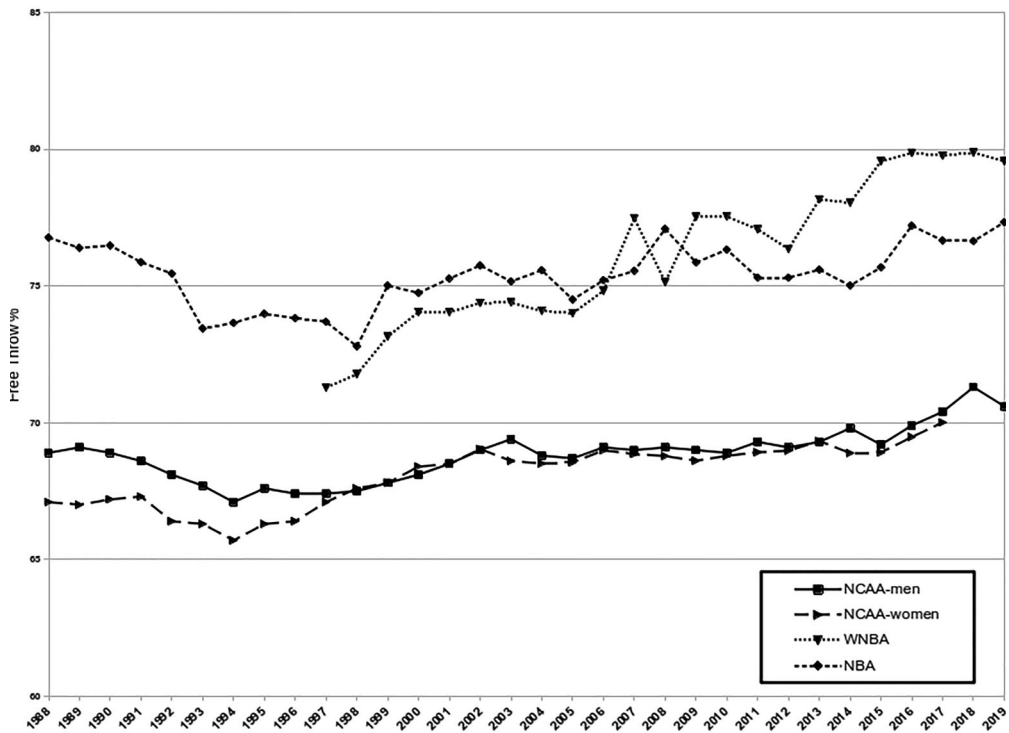


Figure 1. Free-Throw Shooting Performance in US professional basketball (NBA and WNBA) and Division I college basketball (men and women) from 1988 to 2019 seasons. The WNBA started play in 1997. Data are not available for NCAA women for the 2017–2019 seasons. Retrieved from <https://www.basketball-reference.com>

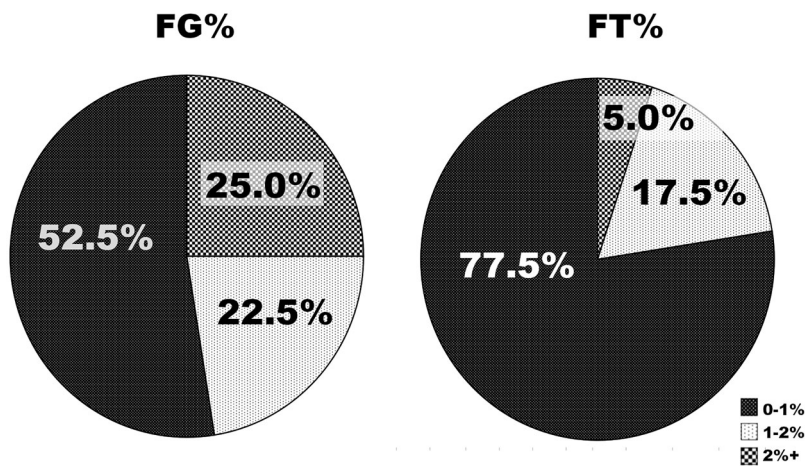


Figure 2. Percentage NBA shooting performance change from season to season (1988–2019). Free-throw performance (FT%) is considerably more stable than field-goal shooting (FG%).

losing teams. Malarranha et al. (2013) found the free-throw rate in the last five minutes of play separated losers and winners also in an international play and lastly, Gómez et al. (2015) showed that during overtime, successful FTS were associated with winning the game.

These findings underscore the importance of FTS in basketball and suggest that late in the game when the score is close, trailing teams often attempt to foul their opponents in order to stop the game clock and challenge them to shoot FTS. This risky strategy is undertaken in the hope that the intense pressure hampers FTS performance and allow the trailing team an opportunity to overturn the score. When the intentional fouling is targeted against players who are known as poor FTS performers it is known colloquially as the “Hack-a-Shaq” (Skinner, 2011) after the notoriously below-average FTS NBA centre Shaquille O’Neal.

Lorenzo et al. (2019) studied the performance of 22 elite European basketball players with at least 10 seasons of first division play. Unlike most game-related statistics, FTS percentages (73% of players in the sample) improved with time. The researchers hence recommended that coaches utilise the most experienced players in the final moments of the game to counter the “Hack-a-Shaq” strategy. Gooding and Gardner (2009) found a similar improvement trend with seniority among NCAA players and Peynircioğlu et al. (2000) among non-experts.

The goal of the current critical review is to examine the many years of research into the psychological foundations of the FTS in order to explicate the reasons behind success and failure on this well-rehearsed task. More specifically we critically screened (i.e., assessed the general interpretation of the results in the context of other evidence, outlined any limitations of the evidence, and the implications of the results for practice as well as future research) and narratively summarised the available research on the proximate socio-psychological underpinnings influencing FTS success. Initially, we provided an overview of archival performance data of the FTS and argued why FTS are of importance in winning a basketball game. We now utilise critical review methodology to answer the overarching question of what situational and individual factors influence performance on FTS. More specifically, the review focused on the influence of pressure, game location (home vs. away), and additional proximate contextual variables that have been investigated in primary FTS research.

Method

Given the broad question addressed and diversity of studies covered in the present review (e.g., self and social identity, social cognition, prejudice and bias, group behaviour and sport research), a critical narrative approach was adopted following the guidelines of Baumeister and Leary (1997; see also Furley & Goldschmied, 2021; Grant & Booth, 2009; Greenhalgh et al., 2018; MacLure, 2005; Saunders & Rojon, 2011). The goal of this approach was to enhance understanding of FTS and theory development (see also Dixon-Woods et al., 2006) by linking together many studies from diverse fields and topics for the purpose of integration or reinterpretation. In contrast to systematic review methodology, this review approach does not require the same predefined, algorithmic search methods with fixed inclusion/exclusion criteria as recommended in the clinical field to answer a narrow question (Higgins et al., 2019). According to Greenhalgh et al. (2018): “the narrative review is a different and potentially complementary form of

scholarship [to the systematic review approach]" (p. 4; see also Greenhalgh, 2012; Ioannidis, 2016; MacLure, 2005 for similar lines of argumentation). Considering calls for a renewed focus on theory development in psychological science (Berkman & Wilson, 2021; Eronen & Bringmann, 2021; Muthukrishna & Henrich, 2019) we consider this narrative approach the more suitable methodology to achieve our overarching goal of advancing understanding of what factors influence performance on FTS.

The presented review approach was targeted to benefit three distinct groups of readers: (1) sport researchers who focus on FTS success as their main target of investigation; (2) basketball practitioners (e.g., coaches, players, and applied sport psychologists) as well as (3) social psychology researchers who employ the FTS, a well standardised task, in their pursuit of understanding more general psychological constructs not specifically associated with basketball.

The literature search of empirical articles published before June 2020 was conducted on Web of Science with the search terms "free-throw shooting", "free-throw performance" also combined with "pressure", "choke/ing", "threat", "anxiety" and "stress". We continued to explore the references of relevant studies until we exhausted the pool. All articles that contained quantitative assessments of performance were included regardless of expertise or sex.

As FTS performance intervention research (largely with a clinical psychology emphasis) as well as shot biomechanics were not the focus of this review, we excluded these type of studies. It is important also to note that while we utilised a social psychology perspective with a focus on performance, we grounded our investigation in proximate motivational factors. However, distal forces which are more remote of which performers may or may not be aware of potentially operate as well. For example, some research has indicated that FTS success may be linked to the perception of flagrant fairness violations. In a case in point, Axt and Oishi (2016) demonstrated in the NBA that when a clear path foul was committed (i.e., when a defender unfairly, usually from behind, hinders an easy scoring opening), FTS success rose above all other type of infractions (including technical or flagrant ones). Haynes and Gilovich (2010) demonstrated that following a blatantly incorrect foul calls, NBA Players made substantially lower percentage FTS but only when the shooter's team was ahead. All FTS articles were read independently by two authors before final exclusion (all articles included which met the criteria). They were reread by another author subsequently for classifications utilised in the results section which were then reexamined by the first author.

Once all papers fitting the criteria set were identified, each was reread by the first two authors to develop a preliminary synthesis of the findings. The data extracted from papers included information on study design and methodology, the sample under study, the country of origin, and the predictor variables under investigation, as well as the results of each study.

We subsequently developed a taxonomy based on major research themes and attempted to describe in the manuscript succinctly the sample, methodology (emphasising strengths and weaknesses) and results of each study, then related these effects to other research in the field. We utilised a social psychology framework for the current review in studying factors influencing FTS success under pressure.

A point of emphasis in the current review is the sample size utilised in each study. Some (e.g., Toma, 2017) had millions of observations while other research seems to incorporate only a few shooting attempts by a small number of participants and may lack in

statistical power (i.e., the ability to detect real differences when they are present) (for example, Czech et al., 2004). Understandably, all else being equal, the “more the better” principle should prevail whereupon added weight should be given to studies with larger samples with sufficiently powered research designs. We attempted to report FTS percentages for each studied condition as they are most important to practitioners in the field. This should allow future research attempts to have a rough estimate of the magnitude of the effect and a reasonable estimate of the number of observations required to ensure sufficient power.

Another noteworthy distinction of the research reviewed pertains to the expertise level of the participants. On the one hand, some studies utilised highly skilled NBA professionals (Toma, 2017; Worthy et al., 2009). On the other end of the spectrum, other research employed novices with only limited FTS familiarity as the subject of their investigation (Krendl et al., 2012; Otten, 2009). Usually, research utilising high-level performers is archival/ correlational in nature, while lower-level participants are employed in experimental work, as professionals are hard to recruit for scientific research.

The results section provides a brief synopsis of the FTS studies included in this review. The discussion section is structured according to the overarching question of what situational and personal factors influence performance on FTS with an emphasis on performance pressure. This partition largely relies on the approach of the primary research and what type of participants are involved. Lastly, a general discussion section is presented with both theoretical/ applied inferences drawn from the critical review of the preceding sections. In addition, limitations and knowledge gaps of the current research and future exploration avenues are identified.

Results and discussion

Study characteristics

Country of origin. The 46 studies originated from ten countries: The United States of America (USA; 12), Spain (4), Australia (2), Canada (1) England (2), Germany (1), Turkey (1), France (1), The Netherlands (1) and Portugal (1). Seventeen were on the NBA making them a combined sample from the USA and Canada (while most of the players originated from the USA certainly a sizeable minority of the participants were internationals). One study used a combined sample from North America, Spain, and Portugal. Two studies did not specify their country of origin.

Type of study. Two types of studies were utilised: archival (23) and experimental (21). Two studies implemented both archival and experimental practices.

Gender of participants. Thirty studies included only male participants. Two studies included only female participants. Twelve studies used both male and female participants. Two studies did not specify the gender of their participants.

Level of participants. Participants were of six different skill levels: Professional (22), collegiate/university (9), experienced (1), college student (4), high school student (1) and novice (3). Two studies consisted of both collegiate and professional athletes. One study consisted of both collegiate athletes and novices. One study consisted of both professionals and novices. One study consisted of both collegiate athletes and college students. One study did not specify the level of participants.

Sample size. Twenty studies had a small sample ranging from three to 43. Fourteen studies had a medium sample ranging from 61 to 490. Eleven studies had a large sample ranging from all FTS ($n = 64,698$) from one NBA season to more than two million FTS extracted from 12 seasons of league play. One study did not specify sample size.

Does game pressure impair free-throw shot performance?

The most common psychological phenomenon associated with the inability to execute FTS successfully is choking following the experience of pressure and associated anxiety. Anxiety has many definitions but a general classification offered by the American Psychological Association's Encyclopedia of Psychology as one of "an emotion characterised by feelings of tension, worried thoughts and physical changes" (Kowalski, 2000). Baumeister and Showers (1986) defined choking as "the occurrence of suboptimal performance under pressure conditions, with the implicit additional assumption that the individual receives utility from a successful performance" (p. 362). Quite similarly, Worthy et al. (2009) defined choking as occurring "when someone underperforms on a task relative to their normal performance because of an acute stressor" (p. 53). In both definitions, the process is characterised as a critical deterioration in the execution of routinely practised tasks due to an elevation in anxiety levels under perceived pressure when optimal performance is sought.

Since FTS is a closed skill with ample time to execute, it is tempting to assume that it is an easy task with a reduced choking probability but since the game comes to a complete standstill before, a substantial increase in attention both by the performer (i.e., heightened self-focus [Baumeister, 1984; Dandy et al., 2001]) as well as others in view (i.e., players, coaches, referees and primarily the crowd), places a significant psychological burden on the performer which is fundamentally different than any other shooting attempt in basketball. Players often describe the feeling as one in which everything else on the court recedes to the background and all attention is focused on the player who is about to shoot. NBA player, J. J. Redick, a superb FTS performer described the experience as, "the only time in a basketball game that everything stops and everyone is looking at you" (Appolonia, 2017). In such a fast-paced team sport with intermittent stoppages, this experience is unique.

Yet, the hiatus in play is not always undermining. It may also depend on the time duration and an extended one may actually assist the FTS performer. Kozar et al. (1993) studied "icing" in the NCAA whereupon opposing coaches request a time-out just before or between crucial FTS supposedly to allow the FTS shooter extra time to ruminate over the consequences of failure. The researchers found that during the last 5 min of games decided by 10 points or less, contrary to their hypothesis, that iced players performed marginally better with a shooting accuracy of 73.4% (350 attempts) than non-iced players (4138 attempts, $M = 68.7\%$). It is plausible that the addition of time actually assisted the players in recovering physiologically or possibly lessen the intense focus felt.

Another impediment for FTS performers, especially when the game is on the line, are high expectations for success since its overall success rate is significantly higher than other shots. TV broadcasters often refer to the FTS line as the charity stripe, highlighting these high expectations (Charity Stripe, 2007). "They call them free for a reason," noted

Mark Price, who is one of only three NBA players with a career FTS percentage over 90% (Fixler, 2012).

In light of the importance of FTS and the lack of historical improvement, much psychological research has focused on understanding the mental duress associated with the shot and finding ways to improve this unique skill. Some studies (Krendl et al., 2012; Oudejans & Pijpers, 2009; Wilson et al., 2009) are experimental in nature and thus allow to infer causality (i.e., internal validity) but are usually limited in their ability to generalise the performance to real game performance as they are often conducted in artificial laboratory settings. Other studies (Baumeister & Steinhilber, 1984; Kozar et al., 1993; Schlenker et al., 1995; Toma, 2017) represent observational and correlational explorations (focusing on data derived from real games). Conversely, this approach tends to be high on external validity (i.e., generalisability) but at the same time suffers in the ability to make clear and unambiguous interpretations of causality. We start with the latter.

Game performance archival studies. The big-data era of sports analytics allowed for the mining of extensive data sets that have been recorded for many years, and are often publicly available for advanced analysis (Morgulev et al., 2018). Toma (2017) utilised this approach to study choking, drawing on over two million observations of FTS. She found that in the final 30 s of tight games, WNBA and NBA players were 5.81 and 3.11 percentage points, respectively less likely to make a FTS. Likewise, the shooting of NCAA men decreased by 2.09 percentage points. Only NCAA women managed to maintain performance under pressure but altogether women and men did not differ in their propensity to choke. Finally and quite surprisingly, the FTS performance of NCAA players who went on to play in the NBA fell 6 percentage points more in high-pressure situations than other players who did not make the professional ranks, potentially because the former had more at stake. One major weakness in Toma's extensive between-subjects research pertains to her conclusion of FTS deterioration under pressure in the end of close games as it is a well-known practice to foul those who are especially sub-par FTS players ("Hack-a-Shaq" strategy) so the comparison may entail very different type of FTS performers.

Worthy et al. (2009) also studied anxiety and choking in close NBA end of game situations but compared within player performance and thus partially addressed Toma's research shortcomings. They studied all regular-season and playoff games for three seasons and examined every FTS by players in the final minute of games within 5 points difference. The researchers showed that players shot worse than their own career average when their team was behind by one or two points or when their team was ahead by one point. However, players' shooting accuracy was not worse when the game was tied. Worthy et al. contended that when one's team was behind, avoidance of losing was most salient and players choked but had a harder time explaining choking when being ahead. We contend that one way to explain subpar FTS performance while in the lead is to study game rules rather than merely motivational states. When ahead with little time remaining some prefer to miss intentionally so that the opposing team would need to go the length of the court to score while a made FTS allows the opposition to inbound the ball from midcourt setting presumably a much better scoring opportunity.

Cao et al. (2011) utilising an extensive NBA data-base while accounting for individual differences in performance, found similarly that players choke under pressure as they

shoot 5–10% worse than normal in the ticking seconds of close games (more pronounced when their team is trailing). Moreover, the researchers found that this deterioration in performance was more apparent for players who had a worse overall FTS success. Similar to Worthy et al. (2009), when games were tied in the final 15 s, no choking effects were detected.

Player type. Gómez et al. (2018) studied 104 games from the top men's Spanish league with a score differential of 2 points or less during the final minute and assessed success in the last pair of FTS (in comparison to career performance). Overall, centres ($M_{\text{FTS}} = 69.4\%$) had the lowest performance compared to guards ($M_{\text{FTS}} = 81.6\%$) and forwards ($M_{\text{FTS}} = 81.3\%$). Moreover, under duress centres deteriorated markedly ($M_{\text{FTS}} = \sim 54.2\%$), guards some ($M_{\text{FTS}} = \sim 73.75\%$) and forwards none ($M_{\text{FTS}} = \sim 80.5\%$). The researchers postulated that since centres shoot closer and often not facing basket fully (e.g., hook-shoot), the FTS is a “difficult” task for them while other positional players tend to shoot from afar with shoulder squared to the basket and hence FTS are more in-line with their regular routine. In high-pressure situations, the performance gap is magnified.

Solomonov et al. (2015) distinguished between NBA players known as clutch (i.e., perceived to improve their performance when the game is on the line) vs. others based on selections made by experts. They compared the performance of the players in the last 5 min of tied games ($n = 222$, during the 2005–2006 season including playoffs) to the last 5 min at the end of the second quarter in the same game. The 16 clutch players identified launched 1000 while others shot 1494 FTS. Clutch perceived players were more likely to possess the ball when the game was on the line and thus were more likely to shoot FTS when fouled and make more shots. However, just like non-clutch players, their FTS efficiency (% makes) at the end of the game, remained unaffected.

Social facilitation: home vs. away analysis

It is true that basketball fans jeer their opponents on a regular basis (Greer, 1983) but the unique delay before and during FTS, allows the local crowd to target their derision at the individual away player (Butler et al., 2018). Hometown crowds try to destabilise shooters of the opposing team by waving varied objects (e.g., towels, pompoms, balloons, foam sticks) and jeer especially hard. The fans sometimes use rhythmic chants in addition to general booing. Recently, fans of certain teams developed elaborate and often surprising plots resembling short theatre acts (Engber, 2015). However, Stone et al. (2008) in a carefully controlled laboratory experiment demonstrated that FTS accuracy was unaffected by projected background displays showing motion (in unison or not) or emotion evoking distractors. In line with this experiment, it is routinely found that overall FTS accuracy by away teams in the NBA is not different from that of the home team (Beuoy, 2016). In the same vein, Epting et al. (2011) subjected Division II College basketball players to a crowd of 10 students who cheered, jeered or stood in silence while performing 10 FTS. Their mean performance ($M_{\text{FTS}} = 82.14\%$) was similar across conditions. Cao et al. (2011) in the NBA failed to detect impaired performance under pressure by venue (home vs. away), regular season vs. playoff games or attendance.

Baumeister and Steinhilber (1984) also focused on the locale and argued counter-intuitively that when one's self-identity was about to be remade (i.e., winning an NBA title) performance decrements would arise in front of supporting crowds due to heightened self-

focus (i.e., the home choke). Studying NBA teams from 1967 to 1982, they showed that FTS success was not different for home and away teams in the first four games of the NBA championship series (72 and 73%, respectively). In the last critical game, though, the home team success declined significantly to 69% while the away team scored at 74%.

Eleven years after the original publication, Schlenker et al. (1995) exploring an expanded data set of NBA seasons from 1967 to 1993, found no statistical differences between home and away teams in the early stages of the series (74% each) and in the last crowning game (73% and 75%, respectively) thus largely debunking the home choke hypothesis.

However, the most recent study (Goldman & Rao, 2012) to explore FTS based on home vs. away settings (though not targeting championship games specifically) used a dataset of about 360,000 NBA FTS. It determined that the performance of home players was significantly worse in clutch situations (especially for poor shooters), while away players did not deteriorate in their ability to score from the line. The researchers argued that the latter finding demonstrated once again the futility of fans' efforts to distract the shooter.

Contextual factors facilitating or debilitating performance

FTS is unique in that the shots are usually performed in the sequence of two (one point each) with exceptions of one or three attempts at times (when the fouled shooting attempt from the field was successful or the failed attempt was beyond the three-point line, respectively). This distinct shooting tandem allows for the temporal study of the relationship between early and later attempts without other game intervening events in between. Studies mostly fail to support a momentum effect (i.e., the hot hand phenomenon) in FTS (Gilovich et al., 1985 [Study 3]; Morgulev et al., 2020). The general pattern found is that of improvement with subsequent shots for the same infraction (Arkes, 2010; Chang, 2019; Phatak et al., 2020) probably as a function of recalibration (i.e., "rescaling of the perceptual-motor system to informational variables", Brand & de Oliveira, 2017, p. 54). This trend has implications to the practice of FTS. Kozar et al. (1995) compared FTS performance of NCAA players in both settings. The researchers highlighted the disparity between the number of shots attempted in sequence during game time (e.g., mostly in clusters of two) in contrast to the many consecutive attempts often undertaken by players during practice. They demonstrated that the accuracy of FTS in games ($M_{\text{FTS}} = 69.2\%$) was not statistically different than the first two attempts in practice ($M_{\text{FTS}} = 69.8\%$) but that additional practice attempts were much more successful ($M_{\text{FTS}} = 76.6\%$) than game performance. It may be the case especially with notoriously inefficient FTS shooters, as in the case of NBA player Dwight Howard who, during the 2012–3 NBA season, was recorded to make 82% of his attempts in practice but converted only 49.2% successfully in games (Freeman, 2013). Maintaining a pre-performance FTS routine (Lonsdale & Tam, 2008; Wrisberg & Pein, 1992) seems to enhance performance overall.

Laboratory studies Mimicking Free Throw Shot Pressure. While archival/ observational research is of great importance in establishing the effect as performance is studied in its original environment, the ability to deduce causality is restricted since the elements of (1) active manipulation and (2) random assignment are missing (and hence alternative explanations for interpretation are possible). Experiments thus are of great importance in

establishing a clear cause-and-effect chain. In addition, laboratory work allows often to explore the stress response of the participants before or while undertaking FTS whereas in archival investigation, heightened pressure is merely presumed based on the score and time remaining. Another point to consider when contrasting these two types of research designs is the level of importance the participants attribute to being successful. Professional players aspire to succeed to sustain their livelihood, stature and self-esteem while laboratory participants despite reinforcement may not be similarly motivated to succeed.

In reviewing the studies in this sub-section it is important to highlight the differences in the operationalisation of the pressure factors believed to influence FTS performance. Some mimic closely stressors and rewards that are present in real game situations (e.g., being financially compensated for successful execution) while others are fairly removed (e.g., reading text prior to shooting). As pressure has been shown to affect perceptual-motor skills differently as a function of skill level (e.g., Beilock & Carr, 2001) we considered expertise in the ordering of the studies below.

Higher-Skilled participants. Oudejans and Pijpers (2009) utilised second division Dutch players and made an effort to simulate game pressure in practice. The participants practised either under pressure conditions or not. The experimental condition entailed visualisation of a decisive game situation, being observed by the coach and teammates, ostensibly being filmed for later evaluation, and a monetary prize in case of success. Players were also penalised by running sprints and push-ups per missed shot. The participants endured their respective regime for five weeks. During the pretest, FTS performance was worse under stress ($M_{\text{FTS}} = \sim 71\%$) than under the no stress condition ($M_{\text{FTS}} = \sim 76\%$). At test time, following the practice period, only the control group deteriorated under stress while the pressure group managed to maintain its performance.

Englert et al. (2015) had experienced basketball players from the fourth German league transcribe a neutral text on a sheet of paper. They were randomly assigned to either an ego depletion condition in which they were instructed to omit the letters “e” or “n” (most frequent letters in German) or undertake the task verbatim (without any restrictions). The ensuing task was to attempt FTS while listening to statements representing worrisome thoughts over headphones. The researchers found that in accordance with ego-depletion theory (Baumeister et al., 1998), basketball players who had initially undergone the ego-weakening manipulation which required them to exercise self-control (i.e., omit condition) performed worse ($M_{\text{FTS}} = 50.42\%$) in contrast to a base-line performance ($M_{\text{FTS}} = 61.33\%$).

Wilson et al. (2009) utilised ten male basketball players from university teams in the UK who experienced both a control and threat condition in which the players were informed that their performance would be compared to others, that the best performers would be rewarded and were told their past performance placed them at the bottom 30% of the sample. Accuracy was lower in the threat ($M_{\text{FTS}} = 50.50\%$) than in the control ($M_{\text{FTS}} = 68.60\%$) condition.

Wang et al. (2004) screened their student basketball players and excluded all those who initially scored less than half of their FTS attempts. The participants shot 20 FTS in a low-pressure condition ($M_{\text{FTS}} = 68\%$) when being observed by the research assistant alone. In the pressure condition, which entailed financial reward (i.e., one dollar for each successful shot as well as \$5 for exceeding personal best) as well as being

watched by others and being filmed, performance dropped significantly ($M_{\text{FTS}} = 63\%$). Participants reported a higher state-anxiety under the pressure condition and a hierarchical multiple regression analyses showed that self-conscious athletes were more susceptible to choking under pressure. In addition, somatic trait anxiety was a significant predictor of choking.

Lastly, Mesagno et al. (2009) focused on distraction as a mean to improve FTS performance. They selected three “choking-susceptible” (based on questionnaire responding) experienced female basketball players and utilised consecutive low/high-pressure conditions in a single case design. All players deteriorated in the pressure condition which entailed video-taping, audience presence, and a performance contingent on financial incentives.

Lower-Skilled participants. Krendl et al. (2012) manipulated pressure for Caucasian students who considered their athletic ability to be at least moderately important. In a pre-post research design, the participants were randomly assigned to either a negative stereotype condition (i.e., a video clip of NBA African-American players shooting FTS concluding with the statement “Black players shoot the highest free-throw percentage in the NBA, making 85%.”) or positive stereotype manipulation (i.e., Caucasian players shooting FTS concluding with a similar statement pertaining to whites). A third group watched an unrelated video clip. Some participants were videotaped during shooting while others were not. When participants were exposed to a negative stereotype their FTS performance declined regardless of whether they were videotaped or not. Those who watched the unrelated video had a similar performance as in base-line which is informative as being merely filmed may not yield the pressure researchers had anticipated.

In line, Otten (2009) utilised a student convenience sample ($n = 201$) who attempted 15 FTS and then were instructed to repeat the task but this time while being videotaped (were informed that it was a pressure manipulation). It was found that while being videotaped, participants performed similarly ($M_{\text{FTS}} = 39.87\%$) as in the no pressure condition ($M_{\text{FTS}} = 37.33\%$). Also, through a structural equation model (SEM) analysis it was found that elevated levels of “perceived control” were associated with better performance under stress.

Finally, Mascaret et al. (2016) attempted to validate the Trier Social Stress Test (TSST). This social evaluation test requires participants to deliver a speech in front of evaluators while supposedly being video-recorded. Exposure to the TSST, yielded an elevated stress responses demonstrated both in self-reported measures (e.g., arousal) and instrumentation (e.g., salivary cortisol). Still, performance among novices did not deteriorate following TSST ($M_{\text{FTS}} = 34\%$) in comparison to before exposure ($M_{\text{FTS}} = 31\%$).

General discussion

The goal of the critical review was to examine the many years of research into the social-psychological foundations of the FTS in order to explicate the reasons behind success and failure on this well-rehearsed skill. We adopted a narrative approach to highlight the diverse subfields of psychological research addressing this highly controlled task while empathising the importance FTS in the game of basketball as it has been shown to often separate winners and losers (Gómez et al., 2015; Navarro et al., 2009; Sampaio & Janeira, 2003). Past research has argued that pressure can either disrupt or alternatively

stimulate performance (Baumeister & Showers, 1986; McCarthy et al., 2012). The critical review approach provided converging evidence for theories arguing for performance declines under pressures instead of performance enhancement.

Archival game research has shown rather consistently that when the score was close and the game was nearing its end, FTS performance deteriorated (Cao et al., 2011; Toma, 2017; Worthy et al., 2009). Experimental research is largely in agreement as high-threat situations decrease FTS success among experts (Englert et al., 2015; Wang et al., 2004; Wilson et al., 2009). The several studies employing novices (e.g., Mascret et al., 2016; Otten, 2009) do not show a similar pattern, possibly since the base-rate of success is initially low. Usually, FTS success, absent pressure, is quite high ranging between 60% and 80% for elite performers (Epting et al., 2011; Gooding & Gardner, 2009; Kozar et al., 1995) but only 30–40% for non-experts (Mascret et al., 2016; Otten, 2009; Tauer & Harackiewicz, 2004).

While archival and laboratory studies show similarly the FTS pressure undermining effect among the non-novice, we also found a significant gap between the two research approaches. Understandably archival explorations utilised elite professionals (Baumeister & Steinhilber, 1984; Toma, 2017; Worthy et al., 2009), whereas manipulated research is nearly bereft of this class of players, often studying those with lower game skills (Englert et al., 2015; Oudejans & Pijpers, 2009), limited experience (Wang et al., 2004; Wilson et al., 2009) or mere novices (Krendl et al., 2012; Mascret et al., 2016; Otten, 2009). Most recently, Laby (2020) studied NBA players' FTS performance in a laboratory setting. We recommend this practice if indeed research conclusions are to generalise to experts.

Another weakness of the reviewed experimental FTS research is that frequently several stressors were manipulated at once (Englert et al., 2015; Wang et al., 2004; Wilson et al., 2009) barring definitive conclusions as to which factors brought about the diminished performance. Future research should develop a targeted FTS taxonomy distinguishing between game-irrelevant stressors (e.g., TSST), game-relevant ones (e.g., compensation, audience effects) and those which are physiological in nature (e.g., running as punishment for missed attempts). Careful delineation of the stress source should assist in the development of a comprehensive theoretical model and intervention.

Lastly, in experimental work, the individual performs for him/herself, while in archival research done in the context of game performance the individual executes in the service of the team. This distinction is not a frivolous one. Tauer and Harackiewicz (2004) demonstrated (albeit with non-experts) that both FTS within-individual performance and intrinsic motivation were higher for team over individual competition.

In devising interventions to ameliorate pressure effects on FTS performance, experimenters should be cautious in attempting to generalise from the non and less skilled to the experts as the divide does not manifest itself in FTS success alone. Cleary and Zimmerman (2001) demonstrated that expert players with a minimum of 70% FTS season success set more specific goals, selected more technique-oriented strategies, made more strategy attributions, and demonstrated higher levels of self-efficacy than non-experts (season FTS average of 55% or less) and novices.

Another recommendation we make in regards to FTS future research is to focus on female performers (Emmonds et al., 2019). Performance considerations should be derived from evidence-based research. Given the under emphasis on FTS research of

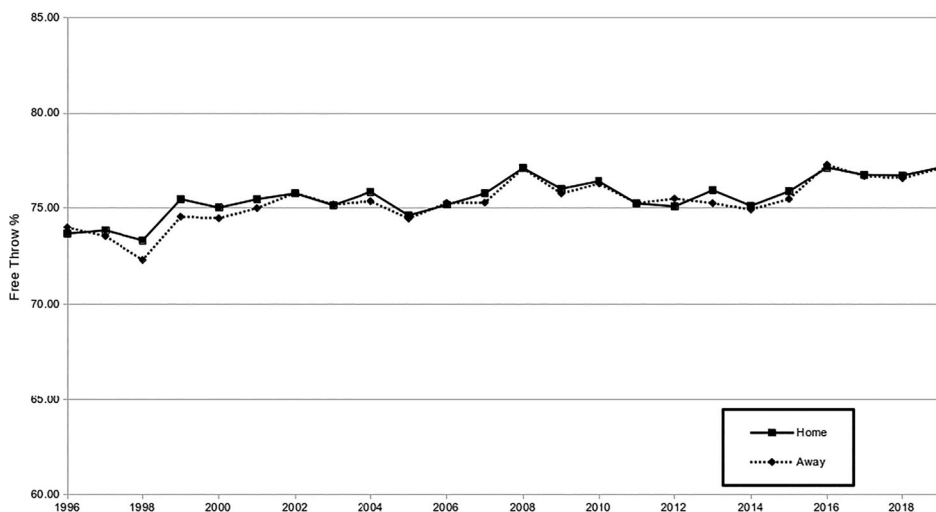


Figure 3. NBA Free Throws Shooting Percentage based on locale (1996–2019 seasons). Data retrieved from www.NBA.com.

elite female athletes with only two studies in our review focused exclusively on women, the current state of affairs of the research may hinder efforts to maximise the potential of elite female athletes.

In summary for FTS experimental research to simulate game pressure, it needs to be most congruent. This pertains to the participants (preference for experts), number of FTS attempted in succession (preferably two) and the type of contest undertaken (team emphasis over individual). This is also true for coaches planning practice. Past research found a discrepancy between superior FTS practice shooting in contrast to diminished game performance (Gooding & Gardner, 2009; Kozar et al., 1995). In line, coaches should be knowledgeable about the importance of proper practice segmentation (Kozar et al., 1995) and the importance of engendering similarity between pressure conditions in practice and games (Oudejans & Pijpers, 2009).

The stoppage of play (in contrast to the continuous nature of the game of basketball itself) and the focus it places on the player before and during execution should be a focus of FTS pressure research. The first type of focus is the one directed by the player to him/herself. Past research (Baumeister & Steinhilber, 1984; Liao & Masters, 2002) argued that this heightened self-awareness state may undermine expert performance because (1) it breaks down well-learned schemas of task execution as the performer now has time to pay careful attention to execution or alternatively, (2) it increases internal locus of attention at the expense of external one which is required for optimal performance. In FTS performance, both Al-Abood et al. (2002) as well as Zachry et al. (2005) showed that giving verbal instructions diverting attention away from body movement to focus on target were successful in enhancing accuracy.

Another type of focus underlined by the stoppage of play is others' attention. We surmise that the undermining pressure effect on FTS execution is largely absent when shooters are facing social facilitation or audience effects. The various operational manipulations of this specific type of influence such as placing a camera for supposedly future

evaluation (Krendl et al., 2012) or altering the crowd's response (Epting et al., 2011) failed to bring about deterioration in performance. Moreover, NBA players seem to perform equally at home and away games (Moskowitz & Wertheim, 2012, p. 118) based on archival research we recorded during more than 20 seasons period (see Figure 3) thus rendering distraction efforts mostly futile (Stone et al., 2008).

The current review, we hope, laid a foundation for a future meta-analysis study to explore the overall effect of pressure on FTS performance in identifying the main moderators in this relationship, namely, skill level, type of stressor utilised and gender. The uniformity of the task as well as the ubiquity of the research should lend itself to this comprehensive statistical undertaking.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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